Structural Persistence Theory: Two-Ledger Accounting, Collapse Modes, and Formal Verification

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Abstract

We present Structural Persistence Theory (SPT), a two-ledger accounting framework for systems that can lose structure and resources in different ways. The multiplicative ledger L measures structural viable-set shrinkage; the additive ledger M measures resource income and cost. Their combination $S_n = M_n \exp(-L_n)$ yields a collapse-mode discriminant: finite-horizon existence failure occurs through the additive resource axis, while the multiplicative structural axis remains positive at every finite step. The structural axis is nevertheless not inert: crossing an L-threshold can change the repair-cost landscape, and recovery is impossible when every target-restoring repair costs more than the available M.

The mathematical foundation is a representation theorem for the structural ledger. Under normalization, additivity, continuity, and nonnegativity, the stage-loss function is forced to be $f(r) = -k \log r$; no non-logarithmic alternative satisfies the axioms. The persistence kernel $m(V_n) = m(V_0) \exp(-L_n)$ is then a telescoping identity. With repair, the net loss $b_t = d_t - r_t$ replaces stage loss, giving $m(V_n) = m(V_0) \exp(-B_n)$. The structural second law—that cumulative total production Σ_n is monotone nondecreasing—is proved as a necessary and sufficient characterization.

The formalization comprises 419 Lean 4 modules with zero sorry/admit/axiom. Conditional bridges to classical results—Shannon's uniqueness theorem, Jaynes' maximum entropy principle, Landauer's principle, the Crooks fluctuation theorem, and others—are constructed as accounting readouts under domain-specific witnesses. The Lean development packages several persistence-facing projections of information theory, thermodynamics, statistical mechanics, dynamics, network science, information thermodynamics, and information geometry through one `MLDomain` interface and a narrow categorical projection. The latest repair-affordability layer also connects native trajectories or certificates—Kalman covariance, BSC retransmission reliability, finite-block achievability/obstruction witnesses and certificates, network repair thresholds, and causal adjustment certificates—to target recovery or recovery infeasibility under a finite resource budget. These results establish a mechanically verified accounting framework and a testable collapse-mode prediction; they do not constitute empirical validation of that prediction.

1 Introduction

1.1 The question

Structure can be lost even when resources remain. An organization may retain budget and personnel yet lose the ability to make coherent decisions. Software may retain computational resources yet become unmaintainable as dependency conflicts accumulate. A long-context language model may retain parameters yet lose logical consistency as unresolved contradictions build up.

In each case, what is lost is not the substrate but the set of states that can still sustain the structure. SPT formalizes this observation: structural loss is the shrinkage of the viable set, measured by a log-ratio scale that is uniquely forced by natural axioms.

1.2 Contributions

1. **Two-ledger accounting**: Additive resources M and multiplicative structural loss L are tracked as distinct ledgers. In the combined scalar $S_n = M_n \exp(-L_n)$, finite-horizon zero crossing is forced by the M-axis, not by the exponential L-axis; the L-axis instead controls structural thresholds and repair-cost lower bounds (§3).

2. **Log-ratio foundation**: Under normalization, additivity, continuity, and nonnegativity, the structural stage-loss function must be $f(r) = -k \log r$. No other form is consistent (§4).

3. **Structural second law**: Cumulative total production Σ_n is monotone nondecreasing, proved as necessary and sufficient under two conditions: positive measure and non-free repair (§5).

4. **Formal verification**: 419 Lean 4 modules, zero sorry/admit/axiom, with conditional bridges to 60+ fields, a theorem map separating formal claims from non-claims, and a bundled `MLDomain` layer for persistence-facing projections of several major theories (§7).

1.3 What this paper does not claim

- SPT does not claim that all systems decay exponentially. The exponential form is a representation theorem for the viable-set measure, not an empirical law about any specific system.

- SPT does not replace domain-specific dynamics. It provides the accounting coordinates; the dynamics are supplied by each domain.

- The conditional bridges are not unconditional proofs of classical theorems. Each bridge requires domain-specific witnesses (choice of viable set V, measure m, and positivity verification).

- The M/L collapse-mode discriminant is a mechanically verified prediction of the accounting model. It is not yet an empirically validated law; the included simulations illustrate the prediction but do not test it against external data.

- The grand-theory layer does not claim that information theory, thermodynamics, statistical mechanics, dynamics, network science, information thermodynamics, and information geometry are the same native theorem. It claims that their persistence-facing projections can be read through the same Lean interface when the required witnesses are supplied.

2 Setup

2.1 Structural maintenance problem

Fix a system X. A *structural maintenance problem* $\Pi = (X, G)$ consists of:

- A state space X.

- A maintenance condition G that defines which states can sustain the structure.

- The *viable set* $V_G = \{x \in X : x \text{ satisfies } G\}$, following Aubin's viability theory [1].

- A *measure* m on X, fixed before observation, that quantifies the "room remaining."

2.2 Viable-set dynamics

Constraints accumulate over time:

$$V_0 \supseteq V_1 \supseteq \cdots \supseteq V_n.$$

Each step consists of *contraction* (constraints shrink V) followed by *repair* (recovery expands V):

$$V_t^- = K_t(V_t), \qquad V_{t+1} = R_t(V_t^-).$$

2.3 Stage loss and repair

The *stage loss* at step t:

$$d_t = -\log \frac{m(V_t^-)}{m(V_t)}.$$

The *repair amount* at step t:

$$r_t = \log \frac{m(V_{t+1})}{m(V_t^-)}.$$

The *net loss*:

$$b_t = d_t - r_t.$$

2.4 Cumulative quantities

Cumulative stage loss: $L_n = \sum_{t<n} d_t$. Cumulative net loss: $B_n = \sum_{t<n} b_t$. Effective resource: M_n (resource-side scalar, defined separately).

2.5 Total production

Under a *repair budget* (repair gain $\leq$ repair cost at each step), total production is:

$$\Sigma_n = B_n + C_n = L_n + \text{slack}_n,$$

where C_n is cumulative repair cost and $\text{slack}_n = C_n - (\text{cumulative gain}) \geq 0$.

3 Two Ledgers and the Collapse-Mode Discriminant

The load-bearing distinction in SPT is the separation of two ledgers:

$$M_{n+1} = M_n + \text{income}_n - \text{cost}_n, \qquad R_n = \exp(-L_n), \qquad S_n = M_n R_n.$$

Here M_n is an additive resource level and R_n is the structural survival fraction induced by cumulative loss. Since $R_n > 0$ at every finite horizon, the sign of the combined persistence scalar is determined by M_n.

Theorem 3.1 (Collapse-mode discriminant). *For $S_n = M_n \exp(-L_n)$ at finite horizon,*

$$S_n \leq 0 \iff M_n \leq 0.$$

Lean: `CollapseModeDiscriminant.collapse_is_additive_not_multiplicative`.

This is the main predictive distinction created by the M/L split. A pure multiplicative structural ledger can decay arbitrarily far but does not hit zero in finite time. Finite-horizon existence collapse, in this model, is an additive resource zero crossing. Lean also proves that the death horizon is independent of the structural decay rate under fixed resource accounting (`death_horizon_independent_of_decay`).

Prediction 3.2 (Operational collapse-mode prediction). *If a system admits separately observable proxies for additive resource balance M_n and multiplicative structural loss L_n, then finite-horizon terminal failure should coincide with an M-axis zero crossing, not with the multiplicative L-axis reaching zero. Structural degradation may precede failure, but the terminal event is predicted to be a resource crossing.*

This prediction is falsifiable in the modest sense relevant here. A single-axis hazard or Kelly-style multiplicative model predicts collapse as loss accumulation on one multiplicative coordinate. The M/L ledger predicts a qualitative separation:

$$\text{gradual structural degradation} \quad + \quad \text{terminal additive resource crossing.}$$

The repository includes a simulation script, `scripts/simulate_collapse_modes.py`, which generates example trajectories and records the predicted event types. The default run produces an M-axis collapse, a mixed-mode M-axis collapse, and a pure L-axis trajectory with no finite zero crossing over the simulated horizon. These are model checks and visualizations, not empirical validation.

The companion network-repair demo, `scripts/simulate_network_repair_affordability.py`, illustrates the repair-feasibility refinement rather than an empirical test. It uses a hand-crafted clustered network, measures L by global-efficiency log loss, and estimates repair cost by a greedy restore-removed-edges policy for a giant-component target. As shown in Figure 1, structural-threshold crossing and the first unaffordable repair step are distinct events. A randomized ensemble companion runs the same separated measurements on ER, scale-free, and small-world graphs under random and targeted edge attacks; it is generated by `scripts/simulate_network_repair_ensemble.py` and writes detail/summary CSV files plus a figure under `data/simulations/` and `paper/figures/`. A small exact-search check, `scripts/validate_network_repair_greedy.py`, compares the greedy repair estimate against brute-force subset search on small random graphs and writes `data/simulations/network_repair_greedy_validation.csv`.

3.1 Irreversibility as repair infeasibility

The collapse-mode discriminant locates the zero crossing, but it is not yet a complete irreversibility criterion. A structure can be arbitrarily degraded without being literally zero, and a resource account can be zero without being absorbing in an open system. The sharper notion is repair feasibility: irreversibility is not identified with damage, decay, or low function. It is a target-relative feasibility claim. Recovery is impossible only when every action that restores the target exceeds the available resource budget.

Given a target structural level T, a set of repair actions a, a repair cost $\mathrm{cost}(a)$, and the restored level $\mathrm{restored}(a)$, recovery to T is feasible from available resource M when

$$\exists a, \qquad \mathrm{restored}(a) \geq T \quad \text{and} \quad \mathrm{cost}(a) \leq M.$$

It is infeasible when every action that reaches the target costs more than the available resource:

$$\forall a, \quad \mathrm{restored}(a) \geq T \Rightarrow M < \mathrm{cost}(a).$$

Theorem 3.3 (Irreversibility as repair infeasibility). *If every repair action capable of restoring the target costs more than the available resource, then recovery to that target is impossible. Lean:* `ReversibilityCriterion.irreversible_if_repair_cost_exceeds_resource`.

The converse sanity checks are also formalized in `ReversibilityCriterion`. If an affordable target-restoring repair exists, irreversibility cannot be concluded from low structure alone. If $M_n = 0$ but the next step has surplus income, then the additive resource ledger becomes positive

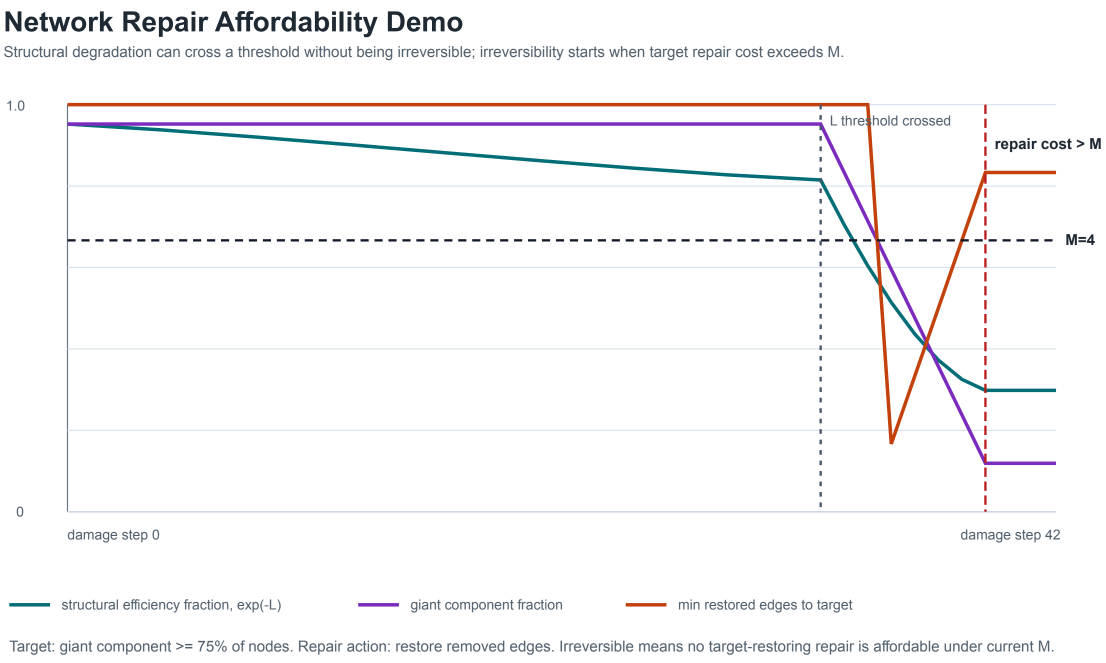

Figure 1: Network repair-affordability demo. The structural efficiency fraction $\exp(-L)$ declines as damage accumulates, and the giant component fragments after bridge removals. The target is a giant component containing at least 75% of nodes; repair cost is the greedy estimate of removed edges that must be restored to meet that target. Threshold crossing occurs before irreversibility: recovery is declared infeasible only when the estimated target-restoring repair cost exceeds the available repair budget M.

again. Conversely, in a closed system, once the resource ledger is nonpositive it remains nonpositive. Thus irreversibility is not "small L" or "zero M" by itself; it is a failed repair-feasibility inequality.

This also gives a precise formulation of irreversible threshold phenomena. An exponential or multiplicative approach to a critical boundary is not, by itself, irreversible. It becomes irreversible when the threshold crossing activates a lower bound on target-restoring repair cost and the available resource lies below that bound:

$$L_n \geq L_{\text{crit}}, \qquad L_n \geq L_{\text{crit}} \Rightarrow \text{cost}(a) \geq C_{\text{crit}}, \qquad M_n < C_{\text{crit}}.$$

Lean: `irreversible_of_threshold_crossing_and_resource_shortfall`.

4 The Representation Theorem

4.1 Axioms for stage loss

A *stage-loss functional* $f\colon (0,1] \to \mathbb{R}$ satisfying:

B2 (Normalization): $f(1) = 0$.

B3 (Additivity): $f(r_1 r_2) = f(r_1) + f(r_2)$.

B4 (Continuity): f is continuous.

NN (Nonnegativity): $f(r) \geq 0$ for $r \in (0,1]$.

4.2 Uniqueness

Theorem 4.1 (Representation). *Any stage-loss functional satisfying B2+B3+B4+NN must have*

$$f(r) = -k \log r, \quad k \geq 0.$$

Proof sketch. B3 is the Cauchy functional equation $f(xy) = f(x) + f(y)$. Under B4 (continuity), the unique solution is $f(r) = c \cdot \log r$. B2 is automatic. Nonnegativity on $(0, 1]$ forces $c \leq 0$, giving $f(r) = -k \log r$ with $k \geq 0$. The Lean theorem map records the exact anchor. □ □

Corollary 4.2 (Persistence kernel). *For any positive mass sequence:*

$$m(V_n) = m(V_0) \cdot \exp(-L_n),$$

where $L_n = \sum l_i$. *At* $k = 1$ *(structural nats),* $S = M \exp(-L)$. *Lean formalizes this as the telescoping kernel.*

4.3 Impossibility

Theorem 4.3 (Impossibility). *No non-logarithmic function satisfies B2+B3+B4+NN. Specifically:*

- *Linear* $a(1-r)$ *violates B3.*
- *Quadratic* $a(1-r)^2$ *violates B3.*

Lean: `ImpossibilityTheorem.impossibility_of_non_log`.

4.4 Relation to Shannon

The representation theorem shares the same mathematical skeleton as Shannon's uniqueness theorem for entropy [3]. Both derive from the Cauchy functional equation; both force a logarithmic form. The difference is in the domain: Shannon measures message uncertainty; SPT measures viable-set shrinkage. Lean: `ShannonUniquenessCorollary`. SPT does not claim to be Shannon's theorem; it shares a common mathematical ancestor and records the correspondence (`NonIdentityTheorem`).

5 The Structural Second Law

5.1 Statement

Theorem 5.1 (Structural Second Law). *Under:*

1. *Positive trajectory:* $m(V_t) > 0$ *for all* t.
2. *Repair budget: gain* $\leq$ *cost at each step.*

Cumulative total production Σ_n *is monotone nondecreasing. This is the deterministic structural second law in Lean.*

Theorem 5.2 (Converse). Σ_n *monotone* $\iff$ *stepTotalProduction* ≥ 0 *at every step. Lean:* `ConverseSecondLaw`.

5.2 Three layers

Layer	Statement	Lean module
Deterministic	Σ_n pointwise monotone	`deterministic_second_law`
Stochastic	$\mathbb{E}[\Sigma_n]$ monotone	`stochastic_second_law`
Coarse-graining	Monotonicity transfers	`coarse_second_law`

5.3 Necessity of conditions

Theorem 5.3 (Free Repair Impossibility)**.** *If gain > cost at any step, Σ can decrease. The repair budget is necessary. Lean:* `FreeRepairImpossibility`.

Theorem 5.4 (Minimal Axioms)**.** *The two conditions are minimal and jointly sufficient. Lean:* `MinimalAxiomTheorem`.

6 Complete Scope Closure

6.1 Applicable systems

Every system with $m > 0$ and gain $\leq$ cost admits the unique form $S = M \exp(-L)$. The two conditions are necessary and sufficient. The functional form is unique and the axiom system is minimal.

6.2 Inapplicable systems

Condition violated	Failure mode	Lean proof
$m = 0$	Log-ratio undefined	`ScopeBoundaryTheorem`
$m < 0$	Physically meaningless	`CompleteScopeClosure`
gain > cost	Second law violated	`FreeRepairImpossibility`
Constant mass	$L = 0$ (theory trivial)	`ScopeBoundaryTheorem`

No fifth category exists. Lean: `CompleteScopeClosure.classification_exhaustive`.

7 Lean 4 Formalization

7.1 Scale

419 modules, 3,583 build jobs. sorry = 0, admit = 0, axiom = 0. Lean 4 v4.26.0 + Mathlib v4.26.0. Repository: `https://github.com/karesansui-u/persistence-lean`.

7.2 Architecture

Layer	Modules	Content
0–4	Core kernel	Telescoping, log-uniqueness, resource budget
5–6	SAT/CSP	Finite-horizon Chernoff/KL bounds
7–8	Markov	Foster-Lyapunov, repair chains
9	Route A	Serial reliability, BSC, fatigue
10–11	Unification	CrossClass, StructuralSecondLaw
Tier S	Meta-theorems	Representation, Impossibility
Tier S+	Necessity	FreeRepair, MinimalAxiom, Converse
Tier S++	Completeness	Stability, Duality, Invariance
Tier S+++	Validation	Falsifiability, NonIdentity

7.3 Conditional bridges

Each bridge formalizes the algebraic readout of the SPT kernel in a specific domain. These are not unconditional proofs of classical theorems but conditional accounting correspondences under domain-specific witnesses. Bridges vary in depth: Tier A bridges invoke the core (representation theorem, telescoping kernel, or second law) to derive non-trivial domain conclusions; Tier B bridges provide correct structural correspondences; Tier C bridges record vocabulary mappings. All are sorry-free, but they are not all at the same depth.

60+ fields are connected, including: thermodynamics, quantum mechanics, statistical mechanics, information theory, probability, biology, economics, engineering, and cosmology.

7.4 M/L interface, grand readings, and categorical projection

The newest Lean layer makes the cross-domain claim more precise. The typeclass `MLDomain` packages the native loss certificate for a domain and gives all instances the same readout theorem, `universal_persistence_law`. A bundled structure, `FiveDomainWitnesses`, records five distinct entry modes: convex/log-Bregman absorption, PAC version-space shrinkage, additive Lyapunov cost, stochastic drift income, and serial-reliability threshold firing. These examples are intentionally heterogeneous: some are new inequalities, some are existing domain theorems routed through the ledger, and the Lyapunov example lives on the additive M-axis rather than being another exponential-shrink copy.

A newer repair-affordability spine makes the interface operational rather than merely semantic. Kalman target covariance recovery is tied to a closed-form covariance trajectory, an observation lower bound, and an observation budget. BSC target reliability is tied to a BSC-native retransmission profile $1 - \text{errorRate}^{r+1}$, bounded monotone reliability certificates, redundancy cost, and repair budget. The finite-block layer is two-sided but certificate-mediated: achievability certificates can imply affordable target recovery, while finite obstruction certificates can imply target-recovery infeasibility; optional rate/capacity metadata may be carried, but it does not generate the certificate. The rank-probability substrate now includes generic finite-uniform event ratios, one-step uniform-PMF event masses, binary span-counting, finite span-escape fraction lemmas, span-complement PMF event masses, and a deterministic full-rank escape product with ratio/product bounds and positivity; these are not yet a random-prefix process, sampled-column independence theorem, or random-matrix rank-failure theorem. The causal layer packages adjustment-formula certificates as producer data for causal-effect and information-gain readouts, without deriving do-calculus or identifiability criteria. These are target-relative recovery statements: the domain supplies a native trajectory, certificate, or witness; the M/L ledger decides whether the target-restoring action is affordable.

The broader `GrandTheoryReadings` structure then records persistence-facing projections of seven large theoretical families:

Reading	Native meaning	Persistence-facing projection
Information theory	entropy, mutual information, KL	loss / information divergence
Thermodynamics	possible change and cost	resource and dissipation accounting
Statistical mechanics	micro–macro passage	Jensen defect under coarse-graining
Dynamics	stability, attractors, drift	Lyapunov-style budget
Network science	structural interaction	threshold and fragmentation readouts
Information thermodynamics	information–energy coupling	product composition of ledgers
Information geometry	statistical shape	Bregman / categorical invariance

This is a semantic bundle, not a triumphal identity claim. The corresponding Lean theorem states that all supplied readings inherit the common persistence profile through the same interface. It does not state that the native domain theorems are identical.

Finally, `domainStructuralProblem` forgets the additive M-axis and projects any `MLDomain` into the existing Mathlib-backed category `StructuralMaintenanceProblem`. Gauge-1 isomorphisms in that category preserve cumulative native loss. This categorical connection is deliberately narrow: it checks compatibility with a genuine category, but it is not a claim that all native theories have been merged into one giant category.

8 Three Observability Layers

Layer	Observability	Role
Specification-fixed	V, m, boundary fixed	Theorems, bounds
Conditional embedding	Drift mapped to SPT	Formal bridges
Structural estimation	Proxy indicators	Prediction, diagnosis

These are not three different theories. They are three observability levels of the same kernel $S = M\exp(-L)$.

9 Related Work

SPT relates to but is structurally distinct from:

- **Thermodynamics**: SPT borrows the accounting style of production, cost, and irreversibility, but it is not a thermodynamic law. It allows open-system repair and resource income, so B_n can be negative even when total production remains budgeted.
- **Information theory**: SPT shares the logarithmic functional form with Shannon-style uniqueness arguments [3], but it measures viable-set shrinkage rather than message uncertainty. The overlap is mathematical, not an identity of domains.
- **Survival analysis and hazard models**: In loss-only mode, SPT coincides with cumulative hazard. The two-ledger extension is different: survival-style models usually place ruin on a single multiplicative axis, whereas SPT predicts finite-horizon existence collapse through the additive resource axis.
- **Kelly and multiplicative growth**: Kelly-style wealth dynamics are multiplicative. SPT can represent multiplicative structural degradation, but its collapse-mode discriminant says terminal existence failure requires an additive resource crossing. This is the key empirical distinction.
- **Viability theory** [1]: SPT adds an accounting layer—measuring how much the viable kernel has shrunk, not just whether viable trajectories exist.

10 Limitations

- **No dynamics**: SPT accounts for structural loss but does not supply dynamical equations.
- **G is external**: The choice of maintenance condition requires domain knowledge.
- **Finite horizon**: Core results are finite-horizon. Asymptotic extensions require additional assumptions.
- **Bridges are conditional**: Each domain bridge requires domain-specific witnesses.

- **Prediction not yet validated**: The collapse-mode discriminant is a formal prediction of the M/L accounting model. External experiments or observational datasets are still needed to test whether real systems fail through the predicted resource zero-crossing pattern.
- **Repair-cost model is external**: The irreversibility criterion requires a domain-specific model of repair actions, repair costs, and target structural levels. SPT supplies the feasibility inequality, not the repair technology.
- **Information-theoretic witnesses/certificates, not Shannon coding theorem**: The BSC and finite-block layers connect retransmission profiles and finite-block achievability/obstruction witnesses and certificates to M/L affordability. Binary span-escape lemmas now reach a one-step finite-uniform PMF event-mass bridge, but they do not yet provide a random-prefix process, sampled-column independence theorem, or random-matrix rank-failure probability theorem. They do not formalize an asymptotic Shannon coding theorem, capacity-achieving construction, or strong converse.
- **Causal certificates, not do-calculus**: The adjustment layer can consume an adjustment-formula certificate and read it through M/L first loss. It does not derive SCM identifiability, back-door, front-door, or do-calculus rules.
- **Semantic, not total, unification**: The `MLDomain` and `GrandTheoryReadings` layers unify persistence-facing projections. They do not subsume the full native content of information theory, thermodynamics, statistical mechanics, dynamics, network science, information thermodynamics, or information geometry.

11 Conclusion

Structural Persistence Theory provides a formally verified two-ledger accounting framework for persistence, degradation, repair, and collapse. The central object is not merely the exponential kernel, but the separation between additive resources M and multiplicative structural loss L. That separation yields a finite-horizon collapse-mode discriminant: structural degradation can be gradual while terminal existence failure is triggered by an additive resource zero crossing. Irreversibility is then sharpened from “near zero” language to a repair-feasibility criterion: recovery is impossible when every target-restoring repair costs more than the resources available.

The log-ratio representation theorem supplies the structural ledger: under normalization, additivity, continuity, and nonnegativity, viable-set shrinkage has the unique form $f(r) = -k \log r$. The Cauchy argument is classical; the contribution is its integration into a two-ledger persistence accounting framework, its scope closure, and its machine-checked cross-domain interface.

The Lean 4 formalization with 419 modules and zero sorry/admit provides machine-checked confidence in the mathematical core. The M/L ledger layer adds a verified collapse-mode discriminant, target-relative repair-affordability readouts, and a bundled cross-domain interface. The result is best described as a mechanically verified unifying accounting framework, together with a testable but not yet empirically validated prediction about collapse modes.

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A Repository

Lean formalization: `https://github.com/karesansui-u/persistence-lean`
Paper repository: `https://github.com/karesansui-u/delta-survival-papers`
Build: `lake build Persistence` (Lean 4 v4.26.0 + Mathlib v4.26.0)
419 modules. 3,583 build jobs. sorry = 0. admit = 0. axiom = 0.